

Vibhu Tummallapalli

Charlotte, NC | 919-909-9573 | vibhu.tummallapalli@gmail.com | github.com/VibhuTummallapalli | US Citizen

EDUCATION

University of North Carolina at Charlotte

Charlotte, NC

Master of Science in Artificial Intelligence

Aug 2026 – May 2028 (Expected)

- Focus: Applied AI/ML, MLOps & AI for Financial Services. Relevant coursework: Applied Machine Learning, Applied Artificial Intelligence, Machine Learning Operations (MLOps), Cloud Computing for Data Analysis, AI-Driven Trustworthy Software Development, Advanced Financial Modeling with AI.

East Carolina University

Greenville, NC

Bachelor of Science in Software Engineering

May 2026

- Concentration: Data Science. Relevant coursework: Artificial Intelligence, Big Data Analytics, NLP, Computer Vision.

SKILLS

Languages: Python, C#, Rust, Java, JavaScript, TypeScript, SQL

AI / ML: Agentic & multi-agent systems (CrewAI, LangChain), Model Context Protocol (MCP), Microsoft Agent Framework, LLM application development, structured prompt engineering, LLM evaluation & testing, Anthropic Claude API & Claude Code, GPT-4o & OpenAI Vision, Ollama, PyTorch

Frameworks & Web: .NET Aspire, .NET 8, ASP.NET Core, Blazor, Entity Framework Core, WPF, React, Next.js, Vite, SignalR, JWT

Cloud & Tools: GCP (App Engine, Cloud Run, Pub/Sub), Docker, GitHub Actions, GitLab CI, PostgreSQL, Bash, PowerShell

WORK EXPERIENCE

Undergraduate Research Assistant

Jun 2025 – May 2026

ECU College of Engineering and Technology

Greenville, NC

- Contributed to 3 faculty research projects in health tech and cybersecurity, owning prototype implementation from design through validation.
- Built data ingestion prototypes in Python, C#, and Java that processed external datasets into clean, structured outputs, informed by literature reviews and performance trade-off analyses.
- Maintained research environments (Python venvs, Maven, local LLM integrations) and managed Git repositories supporting reproducibility across the team.

Learning Assistant

Jan 2023 – Jan 2026

Thomas Harriot College of Arts & Sciences

Greenville, NC

- Tutored 50+ students in Algebra, Calculus, and Statistics, raising failing test scores to 80%+ and helping 75% of students earn a 4.0 in the course.

PROJECT EXPERIENCE

Metamorphic Testing of Multi-Agent AI Systems

Apr 2026

Research · ECU College of Engineering and Technology

Greenville, NC

- Designed a novel catalog of 9 metamorphic relations to test multi-agent LLM systems where traditional test oracles fail (interchangeability, redundancy invariance, graceful degradation).
- Built a reusable Python framework with multi-dimensional semantic verifiers (token overlap, difflib similarity, length ratios) and CrewAI harnesses across OpenAI and Ollama backends via LangChain, generating equivalence rates and violation reports.

MCP Tutor Agent

Jul 2026

Agentic AI Coursework · East Carolina University

Greenville, NC

- Built a local-first AI tutoring agent that calls external tools over the Model Context Protocol (MCP), orchestrated end to end with .NET Aspire.
- Implemented two MCP servers in C# (stdio and streamable HTTP transports) and wired their tool catalogs into an LLM agent using the Microsoft Agent Framework.
- Streamed responses token-by-token from an ASP.NET Core minimal API to a Blazor Fluent UI chat, running fully on local models via Ollama.

Dental Charting Tool

Jun 2025 – Jan 2026

Applied Research · ECU School of Dental Medicine

Greenville, NC

- Built a WPF (.NET)/C# desktop app integrating GPT-4o Vision for radiographic analysis, FDI tooth-numbering charting, CDT-coded procedure cataloging, and urgency-based appointment scheduling.
- Engineered a 5-step prompt pipeline (observation, chart cross-reference, medical alerts, CDT recommendations, urgency triage) for clinically contextualized output; presented at ECU RCAW and CORAS 2026.

NC Early Mathematics Placement Testing

Aug 2025 – May 2026

Online Testing System · Senior Capstone

Greenville, NC

- Built a full-stack web platform (React 19/Vite, .NET 8, EF Core, PostgreSQL) used by 60 NC high schools and 5,000+ students, with a version-controlled question bank, automated grading, and role-based access (JWT + 2FA).
- Added real-time SignalR proctor monitoring and an async email pipeline (GCP Pub/Sub + Python Cloud Function); deployed to GCP through GitHub Actions CI/CD in a 5-member Agile team.